MIL-T-22309A
3 April 1963
SUPERSEDING
MIL-T-22309(SHIPS)
21 January 1960

MILITARY SPECIFICATION

TELEVISION SYSTEM (HIGH DEFINITION FOR CIC DATA PICKUP, DISTRIBUTION AND DISPLAY)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers all services necessary to produce a high resolution television pick-up camera distribution and display system for the purpose of providing multiple remote displays of normal plotted data as provided by the Mark IV, Mod 2, type plotting boards.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification.
MIL-E-17555 - Electronic and Electrical Equipment Associated Repair Parts, Preparation for Delivery of.

STANDARDS

MILITARY
MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ELECTRONIC INDUSTRIES ASSOCIATION STANDARD
RS-170 - Electronic Performance Standards, Monochrome Television Studio Facilities (Nov. 1957).

(Application for copies should be addressed to Electronic Industries Association, 650 Salmon Tower, 11 West Forty-Second Street, New York 36, N. Y.)

3. REQUIREMENTS

3.1 Objectives. - The principal intent of this specification is to provide suitable guides and performance requirement data to permit the economical production of reliable, high performance equipment that is suitable for the requisite shipboard application. Specification MIL-E-16400 and Standard MIL-STD-108 shall apply to all work covered by this specification except as otherwise specified herein.

3.2 Material. - Material used in the production of this system shall be in accordance with Specification MIL-E-16400, except as otherwise specified herein.

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3.3 Construction. - Construction practices shall be in compliance with the stated design objectives of this specification and representative of the highest quality workmanship as specified in Specification MIL-E-16400.

3.4 General requirements. - The equipment shall meet the requirements of Specification MIL-E-16400, except as otherwise specified herein. The design objectives, as specified in Specification MIL-E-16400 for simplicity, reliability, ease of maintenance and installation consistent with the needs of the Naval service, shall be considered during all phases of the equipment design.

3.4.1 Power source. - The system shall perform in accordance with the applicable requirements of this specification when supplied with an input power source of 115 volts, ± 10 percent, 60 cycles per second (c.p.s.), ± 5 percent, alternating current (a.c.).

3.4.2 Enclosures. - All units of equipment shall be suitably enclosed, protected, and ventilated in accordance with the design objectives of Standard MIL-STD-108.

3.4.3 Shock and vibration. - All units of the system shall be produced in accordance with the shock and vibration requirements of Specification MIL-E-16400 for shipboard electronic and optical equipment except as otherwise specified herein. Particular attention shall be paid to this aspect of the camera unit design and construction due to the requirement for rigid overhead mounting to provide a fixed, coplanar optical relationship between the photoconductive (PC) image converter tube photocathode, the camera lens, and the plotting board.

3.5 System. -

3.5.1 System functions. - The system covered by this specification shall perform the specified functions, consistently and with minimum possible requirement for operator skills and readjustment during sustained periods of operation.

3.5.1.1 It shall provide high fidelity television image reproductions of a standard Navy plotting board typified by the Mark IV, Mod 2 equipment, as multiple monitoring stations throughout a ship.

3.5.1.2 The plotting board area to be picked-up, distributed, and displayed by the system will be 5 feet square, located 10 to 15 feet from the coplanar with the television (TV) camera lens mounting plane.

3.5.1.3 All display units in the system shall provide easily read television image reproductions of plotted alphanumeric data, written on any part of the 5-foot square plotting surface, the minimum height of well formed characters to be 0.5 inch. Legible, clearly defined television image reproduction of plotted characters as specified herein shall be possible when the plotting surface is completely covered by the specified characters, plotted with a standard yellow plotting pencil or "china marker", and the characters or lines of characters are separated by minimum strip space widths of 0.1875 inch.

3.5.1.4 Plot illumination applicable to the requirements of 3.5 shall be the maximum illumination available in the Mark IV, Mod 2 plotting board, with overhead space illumination, measured in the horizontal plane, not to exceed 5-foot candles. Space illumination measurement is established with a standard Weston foot candle meter, or equivalent.

3.5.2 System composition. -

3.5.2.1 The system shall be composed of the following units of equipment which shall be identical to the corresponding units of the AN/SXQ-2(V). Nomenclature used is that assigned to the components of Television System AN/SXQ-2:

(a)	Mount, Television Camera	MT-2088/SXQ-2
(b)	Camera, Television	MX-2667/UX
(c)	Control, Television Camera	C-2739/UX
(d)	Power Supply	PP-2205/UX
(e)	Interconnecting Box	J-1017/UX
(f)	Amplifier, Trigger Pulse	AM-2097/UX
(g)	Interconnecting Box	J-1018/UX
(h)	Amplifier, Video	AM-2098/UX

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(i) Generator, Pulse O-577/UX
(j) Cabinet, Electronic Equipment CY-2543/SXQ-2
(k) Viewer, Television IP-487/UX
(l) Control, Television System C-2740/SXQ-2

3.5.2.2 The system shall be capable of furnishing independent video and synchorizing signals for a maximum of eight television viewers.

3.5.3 Mount, Television Camera MT-2088-SXQ-2. - The television camera mount shall provide an appropriately large, rigid bearing surface to permit attachment to a prepared surface with an adequate size and number of machine screws or bolts to withstand the anticipated shock and vibrational stresses covered in the design objective specified herein, in relation to the weight and mass characteristics of the TV camera unit. The mounting assembly shall be detachable from the TV camera electronics package to obviate damage at the time of installation and to permit easy replacement of the functional part of the unit. Provisions shall be made for manual adjustment of 20 degrees in azimuth (pan) and 20 degrees in elevation (tilt) adjustment of the camera optical axis for the purpose of exact optical alignment at the time of installation. These adjustments shall have rugged, positive locking features that will assure permanent alignment under conditions defined herein.

3.5.4 Camera, Television MX-2667/UX. -

3.5.4.1 The television camera shall use a 1-inch diameter photoconductive image converter tube, and shall be designed for overhead mounting in the combat information center (CIC) space by means of the mount described in 3.5.3.

3.5.4.2 The camera shall contain the video preamplifier, line sweep circuit, sweep loss protection circuit, and other circuits as necessary to accomplish the performance objective. The sweep loss protection circuit shall provide indication of loss of either or both camera sweeps by means of a light on the television system control. Particular attention shall be given in the design of the camera to prevention of microphonics in the picture signal due to normally experienced shipboard vibration as specified in Specification MIL-E-16400. Camera dimensions shall not exceed 8-1/2 inches wide by 8-1/4 inches high by 3-1/2 inches deep, exclusive of lens and lens shield.

3.5.4.3 The TV camera cable shall enter the camera in the most direct fashion feasible, and shall not be unduly exposed in a loop or bend. No excess of cable for maintenance shall be included in the installation planning. A separate 25-foot length of camera cable shall be supplied as a service and maintenance facility.

3.5.4.4 The camera lens mounting shall have a locking clamp to secure the lens in its threaded mounting. A protective heavy gauge metal shield shall be provided to protect the lens barrel from direct blows and general mechanical or optical damage due to normal shipboard maintenance and operating functions.

3.5.5 Control, Television Camera C-2739/UX. -

3.5.5.1 The television camera control unit shall be designed for mounting on pull-out slides to allow service access from the front. It shall provide for operation of the television camera in accordance with all applicable requirements of this specification, when connected to the camera by camera cable lengths of 25 to 500 feet.

3.5.5.2 This unit shall contain the video processing amplifier which shall provide keyed clamping, aperture correction, adjustable blanking insertion, and other processing functions necessary to meet the applicable requirements of this specification. The unit shall also contain the vertical sweep circuit and power supplies necessary for proper operation of the camera (with the exception of the main B+ supply).

3.5.5.3 Two isolated 75-ohm internal impedance white positive video outputs shall be provided with a nominal output level of 1 volt into a 75-ohm line.

3.5.6 Power Supply PP-2205/UX. - The power supply shall be designed to supply the necessary B+ to either the television camera control unit, the trigger pulse amplifiers, or the video distribution amplifiers. It shall be designed for standard relay rack mounting, but shall be constructed in such a manner that normal service operations may be performed from the front of the rack. The power supply output shall be sufficiently well regulated to enable maintenance of specified system performance over the full range of input voltage and frequency specified in 3.4.1.

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3.5.7 Generator, Pulse O-577/UX. -

3.5.7.1 The pulse generator shall be designed for mounting on pull-out slides to allow service access from the front. The pulse generator shall employ a binary counter and delay line control of line pulse timings and width. The master oscillator shall be capable of crystal control, free running and line lock operation.

3.5.7.2 Composite blanking, line drive, and field drive pulse outputs shall be provided. Output amplitude shall be adjustable over the range 3 to 5 volts peak to peak into a 75-ohm line. Pulse waveforms shall conform to Electronic Industries Association (EIA) Standard RS-170 insofar as applicable.

3.5.8 Cabinet, Electronic Equipment CY-2543/SXQ-2. -

3.5.8.1 The television camera control unit, pulse generator, synchronizing pulse distribution unit, video distribution unit, and associated power supplies shall be contained in an equipment cabinet, which shall also contain necessary power distribution facilities and a filtered blower assembly for cooling the equipment.

3.5.8.2 Equipment mounted in the cabinet shall be accessible for service from the front. All cable shall enter at the rear.

3.5.9 Control, Television System C-2740/SXQ-2. -

3.5.9.1 The television system remote control unit shall be a compact functionally designed unit, housing all normal operating and set up controls for the system except for the television viewer (s). A hinged cover, or other equally effective means of protecting the controls, with the exception of the starting switch and its associated pilot light, and the sweep loss indicator light, shall be provided.

3.5.9.2 The television system control box shall be suitably constructed to permit horizontal mounting with control shafts in vertical or horizontal planes, or vertical mounting. A preferable configuration will have dimensions similar to, and not exceed the following package dimensions: 17 inches long, 6 inches high, and 6 inches deep. Control identification marking shall be designed to permit proper orientation for unit installation where the major panel dimensions lie in either vertical or horizontal planes. An alternate installation provision may require the application of a suitable supporting panel for standard rack mounting.

3.5.9.3 The control panel shall be provided with controllable edge lighting.

3.5.10 Interconnecting Box J1017/UX. -

3.5.10.1 The interconnecting Box J1017/UX shall consist of a standard relay rack mounting interconnecting box designed to accommodate up to eight each Amplifier Trigger Pulse AM-2097/UX. The interconnecting box shall contain amplifier heater supplies. Coaxial cable connectors shall be provided for each input and output required by the system.

3.5.10.2 Each Amplifier Trigger Pulse AM-2097/UX shall provide two independent outputs at a nominal level of 4 volts peak to peak at 75-ohms, when supplied with an input of 3.5 to 4.5 volts peak to peak. Output waveforms shall conform with EIA Standard RS-170 insofar as applicable, when the amplifier is supplied with an input which conforms with this standard, where applicable.

3.5.11 Interconnecting Box J-1018/UX. -

3.5.11.1 The Interconnecting Box J-1018/UX shall be similar to Interconnecting Box J-1017/UX in mechanical configuration. It shall accommodate up to eight Amplifier Video AM-2098/UX. The interconnecting box shall contain amplifier heater supplies. Coaxial cable connectors shall be provided for each input and output required by the system.

3.5.11.2 Each Amplifier Video AM-2098/UX shall provide one input at a nominal level of 1 volt at 75-ohms. Amplifier gain shall be variable within the limits of 0.9 to 1.2.

3.5.12 Television Viewer IP 487/UX. -

3.5.12.1 The television viewer shall be equipped with a 21-inch rectangular cathode ray tube. Operating controls shall be kept to a minimum and shall be accessible to the viewer, but shall be suitably protected

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from accidental maladjustment or damage by recessed panel location or similar design. The cabinet design shall anticipate shelf and overhead installation. The interspace between the protective safety glass window and the cathode ray tube face plate shall be sealed in such a manner as to positively exclude dust, and shall generally prevent the accumulation of foreign material on the inner surface of the glass or tube face.

3.5.12.2 The unit shall be designed for access for service from the front, preferably by the use of slide rails to allow the chassis to be withdrawn from the cabinet. Modular construction, that is, separately removable subchassis, shall be employed. Overall construction shall be as rugged as possible consistent with reasonable size and weight. The protective safety glass window shall include a circularly polarized filter.

3.5.12.3 The viewer shall provide a highlight brightness of at least 40-foot lamberts without appreciable loss of resolution. Power supplies including the cathode ray tube high voltage supply shall be regulated so as to maintain the required performance over the specified range of power source variations. The viewer shall be designed to operate from negative line and field drive pulses of 3 to 5 volts peak to peak amplitude, and a white positive video signal of 0.25 to 2.0 volts peak to peak.

3.6 Electrical performance. -

3.6.1 Scanning parameters. - Scanning parameters shall be as follows:

Field/frame rate: 60/30 c.p.s. with 2:1 interlace
Lines/frame: 875
Aspect ratio: 1:1

3.6.2 Resolution. - Resolution shall be as follows:

	Center	Corner
Horizontal or line direction	1,000	700 min. TV lines/picture height.
Vertical or field direction	600	500 min. TV lines/picture height.

3.6.3 Sweep linearity. - Sweep linearity shall be as follows:

Camera line and frame: ±1 percent (EIA method in Standard RS-170 where applicable)
Viewer line and frame: ±1 percent (EIA method in Standard RS-170 where applicable)

3.6.4 Geometry. - No picture element shall be displaced by more than 2 percent of the picture height from its true position (EIA method in Standard RS-170).

3.6.5 Sensitivity. - System sensitivity shall be such that 800 television lines per picture height will be resolved in the line direction when televising the EIA resolution chart (Standard RS-170, where applicable) illuminated to a highlight brightness of 5 foot lamberts using an F:15 camera lens.

3.6.6 Gray scale. - The system shall be capable of reproducing the nine steps of the logarithmic gray scale test chart in EIA Standard RS-170.

3.6.7 Shading and spurious signals. -

3.6.7.1 Shading and other similar spurious signals which are synchronized with the scanning frequencies shall be not greater than ±10 percent of the mean peak white signal amplitude.

3.6.7.2 Spurious signals which are not synchronized with the scanning frequencies shall be at least 40 decibels (db) down relative to the mean peak white signal amplitude.

3.7 Elapsed time measurement devices. -

3.7.1 The total operating time of the system shall be recorded by a suitable elapsed time indicator meeting the design objective of this specification.

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3.7.2 A cartridge or vacuum type electrochemical elapsed time indicator shall be suitably connected and located in the camera to measure total operating time of the image converter tube.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. - The system covered by this specification shall be demonstrated to perform in accordance with the requirements of this specification at the contractor's plant to the cognizant design section of the bureau or agency concerned. All facilities including such test patterns as are necessary to demonstrate performance, but excluding the Mark IV, Mod 2 plotting board, shall be provided by the contractor. If demonstration of performance on the Mark IV, Mod 2 plotting board is deemed necessary by the cognizant design section of the bureau or agency concerned, the plotting board will be furnished by the Government to the contractor.

4.2.1 A list of the tests to be performed and the test methods shall be approved by the cognizant design section of the bureau or agency concerned.

4.3 Production inspection. - Production inspection shall be performed in accordance with Specification MIL-E-16400.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. - Unless otherwise specified by the bureau or agency concerned the equipment, accessories, technical publications, and repair parts shall be preserved and packaged by level A or C; packed by level A, B, or C and marked as specified (see 6.1) in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of preservation and packaging and level of packing required (see 5.1).

Custodians:
Army - EL
Navy - Ships
Air Force - AFSC

Preparing activity:
Navy-Ships
(Project 5820-0029)

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

INSTRUCTIONS

BUDGET BU. NO. 45-R309

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY *(Print name and activity)*	DATE

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